



10532 - Integrating Pebble Drift with Disk Chemistry: A New Framework for Interpreting MIRI Water Lines

Cycle: 5, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

Tracing the evolution of water reservoirs in terrestrial planet-forming regions is essential to our understanding of planet formation. Water is commonly detected in MIRI spectra of more than 200 protoplanetary disks, showing a forest of over 1000 water lines. This line forest spans a wide range of excitation temperatures and densities. However, deciphering the information embedded in these rich water spectra is still in its early stages, with the vast majority of analyses based on 1-dimensional slab models. The water reservoir is a combined result of local chemical evolution and water delivery from the outer disk region. An increasing number of observations suggest that the water traced by MIRI can be significantly affected by pebble growth and drift in disks.

We propose to build a new framework that couples pebble growth and drift with 2D thermo-chemical evolution, enabling a self-consistent treatment of water transport, disk thermal structure, and chemical evolution. Using this framework, we will explore a wide range of disk properties and pebble drift conditions, generating simulated MIRI spectra to compare with observations. In particular, we will systematically and homogeneously extract fundamental diagnostic line ratios from both MIRI and simulated spectra, which can be easily adopted for analyzing a large number of MIRI observations. Our results will pave the way for connecting the observable water layers with mid-plane water reservoirs, and determining the pebble flux, which are critical for understanding the impact of pebble drift in providing solid mass supply and delivery of volatiles to the terrestrial-planet forming region.

OBSERVING DESCRIPTION